"BSAI 6A". Since the document only contains the test prompts, I have used my knowledge base to provide the solutions. +1

I've compiled all the unique questions from the document and provided clear, comprehensive solutions for each below.

Quiz 1 Solutions

1. Given the dataset {10, 20, 30, 40, 50}, calculate the mean, median, and apply Min-Max normalization to the value 30. +1

- **Mean:** The average of the numbers.

$$Mean = \frac{10 + 20 + 30 + 40 +}{5}$$

- **Median:** The middle value in a sorted dataset. Since the data is already sorted and has an odd number of values, the median is **30**.
- **Min-Max Normalization:** This scales data to a fixed range, usually 0 to 1. The formula is:

$$z = \frac{x - \min(x)}{\max(x) - \min(x)}$$

For $x = 30$, $\min(x) = 10$, and $\max(x) = 50$:

$$z = \frac{30 - 10}{50 - 10} = \frac{20}{40} = 0.5$$

2. Describe the structure and working of FP-Tree in frequent pattern mining.

- **Structure:** An FP-Tree (Frequent Pattern Tree) is a compact prefix-tree structure used to store quantitative information about frequent patterns in a database. It consists of a root node (labeled "null"), a set of item prefix subtrees, and a frequent-item header table that links to nodes of the same item.
- **Working:** It works in two passes. First, it scans the database to find the frequencies of individual items and discards infrequent ones. It sorts the remaining items in descending order of frequency. Second, it scans the database again, building the tree transaction by transaction. Items in a transaction are added as branches; if transactions share the same most frequent items, they share the same prefix path in the tree, compressing the data significantly.

3. Differentiate between binary, multi-class, and multi-label classification.

- **Binary:** Categorizes data into exactly two distinct classes (e.g., spam vs. not spam).
- **Multi-class:** Categorizes data into three or more classes, but each instance belongs to *only one* class (e.g., classifying an animal as a cat, dog, or bird).
- **Multi-label:** Categorizes data into multiple classes where an instance can belong to *more than one* class simultaneously (e.g., a movie being tagged as both "Action" and "Comedy").

4. Explain the concept of Support Vector Machines (SVM), including hyperplane, margin, and support vectors.

- **Hyperplane:** The decision boundary that separates different classes of data points in an N-dimensional space.
- **Margin:** The distance between the hyperplane and the closest data points from either class. SVM aims to maximize this margin to improve model generalization.
- **Support Vectors:** The data points that are closest to the hyperplane. They are the critical elements of the dataset because they dictate the position and orientation of the hyperplane.

5. Explain the importance of model evaluation techniques in supervised classification.

Model evaluation is crucial because it tells us how well a trained model will perform on unseen, real-world data. It helps identify issues like overfitting (memorizing the training data) or underfitting (failing to learn patterns). Using metrics like accuracy, precision, recall, and the F1-score ensures we choose the most reliable model for deployment.

Quiz 2 Solutions

1. Define Data Mining and explain the Knowledge Discovery in Databases (KDD) process.

Data Mining is the computational process of discovering hidden patterns, anomalies, and relationships within large datasets to extract actionable insights. It is often referred to as KDD (Knowledge Discovery in Databases) because data mining is the core analytical step within a broader, multi-step knowledge discovery framework. +1

2. Describe the major steps involved in the Data Mining / KDD process.

- **Data Cleaning:** Removing noise and handling missing data.
- **Data Integration:** Combining data from multiple sources.
- **Data Selection:** Retrieving relevant data for the analysis.
- **Data Transformation:** Normalizing and aggregating data into appropriate formats.
- **Data Mining:** Applying intelligent algorithms to extract patterns.
- **Pattern Evaluation:** Identifying truly interesting patterns based on usefulness metrics.
- **Knowledge Presentation:** Visualizing and reporting the discovered knowledge to users.

3. Differentiate between Data Mining and simple query processing or expert systems. +1

- **Simple Query Processing:** Retrieves explicit, existing facts from a database (e.g., "Show me all users who bought shoes today").
- **Expert Systems:** Rely on hard-coded rules manually programmed by human experts.
- **Data Mining:** Automatically discovers *new, hidden* rules and predictive patterns from data without being explicitly programmed with those rules (e.g., "Predict which users are 80% likely to buy shoes next week").

4. Explain the role of data preprocessing in the data mining process.

Real-world data is often incomplete, noisy, and inconsistent. Data preprocessing (which includes cleaning, integration, reduction, and transformation) resolves these issues. High-quality data is essential because "garbage in" leads to "garbage out"—preprocessing ensures the mining algorithms produce accurate and reliable results. +1

5. A bank wants to predict loan approval status based on customer attributes. Identify the type of classification problem and suggest suitable algorithms.

- **Type:** This is a **Binary Classification** problem (the outcomes are either "Approved" or "Rejected").
- **Algorithms:** * *Logistic Regression:* Great for binary probabilities and highly interpretable.
  - *Decision Trees / Random Forest:* Excellent for banking because the decision paths (e.g., "If income > 50k and credit score > 700") are easy to explain to customers and regulatory bodies.

6. Describe common data mining techniques (Decision Trees, Naïve Bayes, KNN, SVM, Apriori Algorithm).

- **Decision Trees:** Uses a tree-like model of decisions and their possible consequences to classify data.
- **Naïve Bayes:** A probabilistic classifier based on applying Bayes' theorem with an assumption of independence between features.
- **KNN (K-Nearest Neighbors):** Classifies a data point based on how its closest neighbors (data points) are classified.
- **SVM:** Finds the optimal hyperplane to separate data into classes.
- **Apriori:** Used for association rule learning to find frequent itemsets (e.g., "Customers who buy bread also buy butter").

7. Why has data mining become important in the era of digital data explosion? Discuss the importance of presentation and visualization.

We live in an era where massive volumes of data are generated daily, but humans cannot manually process it to find value. Data mining is essential to convert this raw data into profitable information. Visualization and presentation are equally critical because they translate complex algorithmic outputs into intuitive graphs and charts, allowing non-technical business leaders to understand and act on the findings.

+2

Quiz 3 Solutions

1. Explain how data mining is applied in business, healthcare, and finance sectors.

- **Business:** Market basket analysis, customer segmentation, and targeted marketing.
- **Healthcare:** Predicting disease outbreaks, personalized medicine, and evaluating the effectiveness of treatments.
- **Finance:** Credit scoring, algorithmic trading, and forecasting market trends.

2. Discuss the key applications of Data Mining in fraud detection and anomaly detection.

Data mining establishes a baseline of "normal" behavior (e.g., a user's typical spending habits). It then scans massive transaction logs in real-time to flag anomalies—deviations from this norm—such as a credit card suddenly being used in a different country or unusual network traffic indicating a cyberattack.

3. Describe the advantages of data mining in operational efficiency and risk management. +1

- **Operational Efficiency:** It identifies bottlenecks in supply chains, predicts when machinery will need maintenance (preventative maintenance), and optimizes resource allocation to save time and money.
- **Risk Management:** It helps organizations predict the likelihood of adverse events (like a customer defaulting on a loan or a supplier failing to deliver) so that mitigating actions can be taken proactively.

4. Discuss the challenges in data mining, including data quality, privacy, overfitting, and underfitting. +1

- **Data Quality:** Mining algorithms require clean data; noise, missing